Does the Galaxy Acceleration Scale Track Cosmic Dark Energy? A Non-Circular Cross-Scale $a_0(z)$ Test of de Sitter–Unruh Modified Inertia — the $z = 0$ Tie, the High-z BTFR Forecast, and Why $M_{\rm bar}$ Calibration (Not Rotator Count) Is Decisive

Carl Zimmerman Briar Creek Tech · carl@briarcreektech.com

Abstract

In a de Sitter–Unruh modified-inertia (MI) framework the low-acceleration scale a_0 is fixed to the cosmological horizon by $a_0 = cH_\Lambda/Z$ with $Z = \sqrt{32\pi/3} = 5.78881$, giving the pure–dark-energy-density (canonical) footing $a_0^{\rm canon} = c^2\sqrt{\Lambda/32\pi} = 9.355 \times 10^{-11}\,{\rm m\,s^{-2}}$ or the total-density (ALT) footing $a_0^{\rm ALT} = cH_0/Z = 1.1305 \times 10^{-10}\,{\rm m\,s^{-2}}$, two footings 20.9% apart that are carried on every number here. The framework does **not** rewrite Type Ia supernova standardization and gives **no** new distance formula — its cosmological background is General Relativity. Its distinctive content is that the dark-energy term in the Friedmann equation is not a free fit but the *galaxy* acceleration scale: $H^2(z) = \frac{8\pi G}{3}\rho_m(z) + Z^2 a_0^2(z)/c^2$, with $a_0(z)$ measurable from galaxy dynamics and Z a single posited constant. This ties a galaxy-kinematic quantity to a cosmic-distance quantity through $a_0(z) = \frac{c}{2}\sqrt{G\rho_{\rm DE}(z)}$ (equivalently $a_0 = c^2\sqrt{\Lambda_{\rm eff}/32\pi}$ with $\Lambda_{\rm eff} = 8\pi G\rho_{\rm DE}/c^2$), i.e. $a_0(z)/a_0(0) = \sqrt{\rho_{\rm DE}(z)/\rho_{\rm DE,0}}$ — a bridge ΛCDM structurally lacks. It adds no dark-energy parameters beyond the GR background: the same $\rho_{\rm DE}(z)$ that fits cosmic distances must fit the galaxy $a_0(z)$. The equality $a_0 \leftrightarrow \Lambda$ at fixed z is a definition and is circular; the *non-circular* test is the joint z-tracking of two independent datasets. We report a method + forecast **non-detection with a live cosmology-side hint**. (i) The $z = 0$ tie holds: Λ-blind SPARC $a_0(0) = 1.181 \times 10^{-10}\,(\pm 16\%)$ versus cosmic $(c/2)\sqrt{G\rho_{\rm DE,0}}$ gives ratio 1.26 (1.44σ, Planck H_0) or 1.17 (0.95σ, SH0ES H_0). (ii) The DESI DR2 $w_0 w_a$ posterior propagated through the framework relation yields $a_0(z{=}3)/a_0(0) \simeq 0.74$ (Pantheon+ 0.775, DESY5 0.737, Union3 0.707; a $\sim 2.2\sigma$ decline) — a consequence of the framework relation plus the DESI $w(z)$, not an independent detection. (iii) On the galaxy side the distinctive 0.60–0.75 decline is neither detected nor excluded: separability $S \simeq 0.8\sigma$. One clean deep-MOND $z = 3.25$ rotator (the Big Wheel) gives $a_0(3.25)/a_0(0) = 1.31\,(+0.93/-0.52)$, consistent with a constant and excluding the ALT-cH_0 $\sim 5\times$ *rise* at $\sim 2\sigma$, but unable to separate the 0.70 decline from flat. The central finding is methodological: in the baryonic Tully–Fisher (BTFR) estimator a_0 is algebraically 1:1 degenerate with baryonic mass ($\Delta \log a_0 = -\Delta \log M_{\rm bar}$ at fixed V), so the dominant $M_{\rm bar}$ systematic is common-mode and does **not** beat down with rotator count: with

a $\gtrsim$ 0.08-dex correlated $M_{\rm bar}$ floor the test never reaches 3σ at any N. The decisive lever is $M_{\rm bar}$ calibration to $\lesssim$ 0.05 dex common-mode, not the number of galaxies, and the a_0-separable observable that breaks the degeneracy is the full rotation-curve / radial-acceleration transition, not the BTFR zero-point. The value of a_0 and the constant Z are posited, not derived; the a_0 magnitude inherits Z, so only the ratio/tracking is under test.

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1. Introduction

1.1 The framework and the horizon scale

Modified inertia proposes that the inertial response of a body to a force weakens below a critical acceleration a_0, rather than the gravitational field being modified. In the de Sitter–Unruh realization, an accelerated observer in a universe with a cosmological horizon sees a floor in the vacuum's response set by the horizon temperature, and the transition scale is tied to the horizon:

$$a_0 = \frac{cH_\Lambda}{Z} = \frac{c^2\sqrt{\Lambda/3}}{Z}, \qquad Z = \sqrt{\tfrac{32\pi}{3}} = 5.78881,$$

with $H_\Lambda \equiv c\sqrt{\Lambda/3}$ the de Sitter (pure dark-energy) Hubble rate. The Planck 2018 value $\Lambda = 1.089 \times 10^{-52}\,\mathrm{m}^{-2}$ gives the **canonical footing** $a_0^{\rm canon} = c^2\sqrt{\Lambda/32\pi} = 9.355 \times 10^{-11}\,\mathrm{m\,s}^{-2}$ (pure $\rho_{\rm DE}$). A second **ALT footing** replaces the pure-Λ density by the total present-day critical density, $a_0^{\rm ALT} = cH_0/Z = 1.1305 \times 10^{-10}\,\mathrm{m\,s}^{-2}$. The two differ by 20.9%; both are carried on every dimensional number below. Nothing here derives the value of a_0, the constant Z, or the sign of the inertial correction: those remain postulates. What the framework ties to the horizon is the *scale* of a_0.

1.2 The parameter-free replacement of the free-w fit

The framework keeps General Relativity for the cosmological background and therefore has no native supernova formula: the light-curve → distance standardization is unchanged, and the Friedmann equation is the standard one. Its single distinctive move is that the dark-energy term is not a free fit but the galaxy acceleration scale,

$$H^2(z) = \frac{8\pi G}{3}\,\rho_m(z) + \frac{Z^2 a_0^2(z)}{c^2}, \qquad \frac{Z^2 a_0^2}{c^2} = H_\Lambda^2 = \frac{\Lambda c^2}{3} \ \ (z=0),$$

so that, inverting $Z^2 a_0^2/c^2 = \frac{8\pi G}{3}\rho_{\rm DE}$ with $Z^2 = 32\pi/3$, the dark-energy mass density is $\rho_{\rm DE}(z) = 4a_0^2(z)/(Gc^2)$ and

$$\boxed{a_0(z) = \tfrac{c}{2}\sqrt{G\,\rho_{\rm DE}(z)} \;=\; c^2\sqrt{\Lambda_{\rm eff}(z)/32\pi} \quad\Longrightarrow\quad \frac{a_0(z)}{a_0(0)} = \sqrt{\frac{\rho_{\rm DE}(z)}{\rho_{\rm DE,0}}}}$$

The SI form $\frac{c}{2}\sqrt{G\rho_{\rm DE}}$ uses the dark-energy *mass* density $\rho_{\rm DE}$ (kg m^{-3}); the equivalent geometric form uses $\Lambda_{\rm eff} \equiv 8\pi G\rho_{\rm DE}/c^2$ (units m^{-2}). Both give $a_0^{\rm canon} = 9.355 \times 10^{-11}$ at $z = 0$ on Planck $\rho_{\rm DE,0}$; the two conventions must not be mixed ($c^2\sqrt{\rho_{\rm DE}/32\pi}$ with a *mass* density is dimensionally inconsistent). The redshift-tracking ratio $a_0(z)/a_0(0) = \sqrt{\rho_{\rm DE}(z)/\rho_{\rm DE,0}}$ is convention-independent.

On the supernova side this adds **no free dark-energy parameter** — against ΛCDM's Ω_Λ (and, in the evolving-DE extension, w_0, w_a). The framework does not, however, produce a parameter-free dark-energy history from nothing: it *consumes* whatever $\rho_{\rm DE}(z)$ the cosmic data prefer (itself the w_0, w_a fit to distances) and requires that the *same* function reproduce the galaxy-side $a_0(z)$. The content is therefore a cross-*dataset* constraint — one $\rho_{\rm DE}(z)$ must simultaneously fit cosmic distances and galaxy kinematics — not a first-principles prediction of $w(z)$. At $z = 0$ the framework is identical to ΛCDM by construction ($Z^2a_0^2/c^2 = \Lambda c^2/3$). It differs from ΛCDM only if $a_0(z)$ evolves — that is, only if the galaxy acceleration scale changes with redshift in lock-step with the cosmic dark-energy density.

1.3 The non-circular cross-scale idea

ΛCDM gives no reason for a galaxy's internal acceleration scale to track cosmic expansion; the framework forces exactly that link. Crucially, the equality of a galaxy a_0 and a cosmic Λ at a *single* epoch is a definition — one can always convert one into the other, and reading agreement there is circular. The genuine, non-circular test is whether **two independently measured functions of redshift** — a_0 from galaxy kinematics and $\rho_{\rm DE}$ from cosmic distances — track each other as z varies. That z-tracking is a prediction ΛCDM cannot make and the framework cannot avoid. The framework *inherits* the dark-energy history $w(z)$ (it does not predict $w(z)$); its content is the tie between the two datasets, so any decline it exhibits is a consequence of the framework relation combined with the measured $w(z)$, not an independent statement about dark energy.

1.4 Credit for the kernel, and the status of this paper

The low-acceleration interpolation used to read a_0 from rotation curves, $\nu(y) = \sqrt{1 + 1/y}$ with $y = g_{\rm bar}/a_0$, is Milgrom's (1999, *Phys. Lett. A* **253**, 273, Eq. 9), in the original MOND context of Milgrom (1983). Milgrom's coefficient was $2cH_\Lambda$ rather than cH_Λ/Z; the deep-MOND limit $V^4 = a_0\,GM_{\rm bar}$ that anchors the galaxy-side measurement here is the baryonic Tully–Fisher relation (McGaugh 2005, 2012; Lelli, McGaugh & Schombert 2016). The distinctive content the present framework contributes is the cH_Λ/Z coefficient and the $a_0 \leftrightarrow \rho_{\rm DE}(z)$ tie under test. This is a **method + forecast paper reporting a non-detection** with a live cosmology-side hint and a strategic finding; it is not, and does not claim to be, a detection. All load-bearing numbers are produced by committed, exit-0 scripts (Appendix A), and both footings are carried throughout.

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2. The relation and why it is non-circular

Terminology. *Footing* — the choice of density that sets the a_0 *magnitude*: canonical (pure $\rho_{\rm DE}$, 9.36×10^{-11}) or ALT (total present-day density, 1.13×10^{-10}); the two are 20.9% apart and cancel in any ratio. *Deep-MOND* — the low-acceleration regime $g \ll a_0$, where $\nu \to \sqrt{a_0/g}$ and $V^4 = a_0 G M_{\rm bar}$ holds with fitted slope exactly 4. *BTFR zero-point* — the normalization of $V^4 = a_0 G M_{\rm bar}$; in the deep-MOND limit that zero-point *is* a_0, so reading $a_0(z)$ and reading the BTFR zero-point are the same operation. *Common-mode* — a systematic applied identically across a sample (one $\alpha_{\rm CO}$/IMF/dust prescription), which does not average down with sample size N.

2.1 The two-dataset bridge

Write the framework tie as a bridge between measurements made on wholly different physical scales:

$$\underbrace{a_0(z)}_{\text{galaxy kinematics: BTFR zero-point}} = \frac{c}{2}\sqrt{G\,\rho_{\rm DE}(z)} = \underbrace{\frac{c}{2}\sqrt{G\,\rho_{\rm DE}(z)}}_{\text{cosmic distances: SNe / DESI } \rho_{\rm DE}(z)}.$$

The left side is read from the internal dynamics of individual galaxies (parsec–kiloparsec scales); the right side is read from the expansion history (gigaparsec scales). ΛCDM contains no relation forcing these to agree at all, let alone to co-evolve. The framework forces both their $z = 0$ equality and their joint z-dependence.

2.2 Definition versus test

The $z = 0$ equality $a_0(0) = (c/2)\sqrt{G\rho_{\rm DE,0}}$ is definitional: given a cosmic $\rho_{\rm DE,0}$ one *defines* the horizon a_0, so a $z = 0$ match is a consistency knot, not an independent success. Its value is as the anchor the bridge extends from — if it failed badly the whole construction would be dead on arrival — but it carries no evidential weight beyond internal consistency. The non-circular content is entirely in the *slope*: does galaxy-measured $a_0(z)$ decline (or stay flat, or rise) in the same way the independently measured cosmic $\rho_{\mathrm{DE}}(z)$ does? Because the two datasets never share inputs, a wrong background assumption shifts both together and cannot manufacture a spurious co-evolution (quantified in §4.4).

2.3 What the framework predicts, and what it inherits

At fixed z the framework prediction is fully specified once $\rho_{\mathrm{DE}}(z)$ is chosen. Two reference cosmic tracks bracket the question:

track	(w_0, w_a)	$a_0(3)/a_0(0)$	$a_0(3.25)/a_0(0)$
flat-Λ ($w = -1$)	$(-1.00,\ 0.00)$	1.000 (flat)	1.000
DESI-DR2 evolving	$(-0.75,\ -0.90)$	0.712 (declining)	0.685
(ALT $cH(z)$ evolution model, for contrast)	$a_0 \propto cH(z) = cH_0E(z)$	4.57 (rising)	4.99

with CPL history $\rho_{\mathrm{DE}}(z)/\rho_{\mathrm{DE},0} = (1+z)^{3(1+w_0+w_a)}\exp[-3w_a z/(1+z)]$. The framework's distinctive canonical-footing prediction is a **mild decline**, $a_0(3)/a_0(0) \approx 0.60$–$0.75$, if the DESI-preferred evolving dark energy is real. If instead $w \to -1$ (flat), the framework reduces to ΛCDM at all z and the distinctive signal vanishes, leaving only the $z = 0$ tie. The third row is **not** the ALT $z = 0$ normalization ($a_0^{\mathrm{ALT}} = 1.131 \times 10^{-10}$) but a distinct *evolution model* in which a_0 tracks the total-density Hubble rate $cH(z)$ and therefore *rises* steeply — the opposite sign — and it is the one branch the current galaxy data can already address (§4.3). The ALT $z = 0$ normalization itself, if it instead tracked $\rho_{\mathrm{DE}}(z)$, would decline like the canonical footing and is *not* excluded; only the rising $cH(z)$ evolution is. The flat–DESI separation is only 0.147 dex at $z = 3$ (0.164 dex at $z = 3.25$); that small gap is what the galaxy data must resolve.

A theory-side caveat bounds the whole construction. The horizon relation $a_0 = cH_\Lambda/Z$ is derived for a *static* de Sitter horizon (constant Λ). Promoting it to $a_0(z) = \frac{c}{2}\sqrt{G\rho_{\mathrm{DE}}(z)}$ for an evolving equation of state ($w \neq -1$) is an **adiabatic / instantaneous-horizon ansatz** — the assumption that the inertia scale tracks the instantaneous dark-energy density — not a result derived from the de Sitter–Unruh construction, whose horizon is that of a constant-Λ background. The z-tracking under test is therefore a test of this ansatz combined with the measured $w(z)$; a null would disfavour the ansatz, not the static-horizon relation it reduces to at $z = 0$.

3. The $z = 0$ tie and the cosmology side

3.1 The $z = 0$ tie

The Λ-blind galaxy $a_0(0)$ is taken from the gas-dominated SPARC BTFR zero-point measured without any cosmological-Λ input (the companion a_0-line/Λ paper, Zimmerman 2026, DOI 10.5281/zenodo.21419735): $a_0(0) = 1.181 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ ($\pm 16\%$). The cosmic-side prediction is $a_0(0) = (c/2)\sqrt{G\rho_{\mathrm{DE},0}}$ with $\rho_{\mathrm{DE},0} = \Omega_{\mathrm{DE}}\, 3H_0^2/(8\pi G)$:

footing	cosmic $a_0(0)$	$a_{0,\mathrm{SPARC}}/a_{0,\mathrm{cosmic}}$	tension
Planck ($\Omega_{\mathrm{DE}} = 0.685$, $H_0 = 67.4$)	9.36×10^{-11}	1.26	1.44σ
SH0ES-H_0 ($\Omega_{\mathrm{DE}} = 0.685$, $H_0 = 73.0$)	1.01×10^{-10}	1.17	0.95σ

The tie **holds at** $\sim 1\sigma$ on the canonical (Planck-H_0) footing and slightly better (0.95σ) at the local H_0. This is by construction — the definitional anchor — and holds. It re-states that the framework enters the high-z test on solid $z = 0$ footing, and carries no evidential weight beyond that internal consistency.

3.2 The cosmology side: DESI DR2 propagated

On the cosmic side there is no free dark-energy parameter to fit — the framework consumes whatever $\rho_{\mathrm{DE}}(z)$ the data prefer. We propagate the DESI DR2 (2025, arXiv:2503.14738) $w_0 w_a$CDM posterior (BAO + CMB + SNe; representative marginals with w_0–w_a correlation ≈ -0.87; ΛCDM excluded at 2.8σ/Pantheon+, 4.2σ/DESY5, 3.8σ/Union3) through $a_0(z)/a_0(0) = \sqrt{\rho_{\mathrm{DE}}(z)/\rho_{\mathrm{DE},0}}$ (Appendix A, `desi_posterior_a0z.py`):

SNe combination	$a_0(3)/a_0(0)$	$[16\%, 84\%]$	$P(a_0$ declines$)$	decline signif.	in 0.60–0.75
DESI+CMB+Pantheon+	0.775	[0.683, 0.878]	97.9%	2.0σ	38%
DESI+CMB+DESY5	0.737	[0.651, 0.833]	99.3%	2.5σ	51%
DESI+CMB+Union3	0.707	[0.599, 0.831]	98.3%	2.1σ	48%

Across the three combinations the median is $a_0(3)/a_0(0) \simeq 0.74$ with a $\sim 2.2\sigma$ decline, and all three posteriors overlap the framework's 0.60–0.75 band. This is a genuine, *live* cosmology-side hint: **if** the DESI evolving-dark-energy signal is real, the induced $a_0(z)$ decline falls squarely inside the framework's 0.60–0.75 band. It is not, however, an independent test — it is a re-expression of the DESI $w(z)$ result through the framework relation, the *same* DESI information seen twice, and it therefore **cannot be added to the galaxy-side** χ^2 as if it were a second measurement. It dissolves entirely if the DESI signal regresses to $w = -1$. It sharpens the *target* for the galaxy side; it does not by itself confront it.

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4. The galaxy-side $a_0(z)$ from the high-z BTFR

4.1 The estimator

On the galaxy side, $a_0(z)$ is read from the BTFR zero-point at each redshift. In the deep-MOND regime $V^4 = a_0\, GM_{\rm bar}$ is exact (the fitted slope is forced to 4), so at fixed circular velocity

$$\frac{a_0(z)}{a_0(0)} = \frac{(V_z/V_0)^4}{M_{{\rm bar},z}/M_{{\rm bar},0}}, \qquad \text{equivalently} \qquad \Delta \log a_0 = -\,\Delta \log M_{\rm bar}\big|_{\text{fixed } V}.$$

Reading $a_0(z)$ is therefore the same operation as reading the BTFR zero-point. The estimator is valid **only** where the fitted slope is genuinely 4, i.e. for objects deep in the low-acceleration regime $g \ll a_0$.

4.2 The one clean high-z datum: the Big Wheel at $z = 3.25$

A single clean deep-MOND high-z rotator is currently available — the Big Wheel (Wang et al. 2025, *Nature Astronomy* **9**, 710; $z = 3.2514$; arXiv:2409.17956), a large, low-acceleration disk whose outer rotation curve satisfies the deep-MOND condition. Monte-Carlo over its velocity and baryonic-mass systematics ($V_{\rm rot} = 280 \pm 31\,{\rm km\,s^{-1}}$, $\sigma_{\rm int} = 61\,{\rm km\,s^{-1}}$ with asymmetric-drift correction; $M_\star = 1.7 \pm 0.7 \times 10^{11} M_\odot$, dynamically capped; $M_{\rm gas} = 1.8 \pm 0.9 \times 10^{11} M_\odot$, $\times 1.36$ He) gives $a_{0,\rm eff} = 1.54\,(+1.10/-0.61) \times 10^{-10}\,{\rm m\,s^{-2}}$, i.e.

footing	$a_0(3.25)/a_0(0)$
SPARC 1.181×10^{-10}	$1.31\,(+0.93/-0.52)$
canonical 0.936×10^{-10} ($\rho_{\rm DE}/cH_\Lambda$)	$1.65\,(+1.17/-0.65)$
ALT 1.131×10^{-10} ($\rho_{\rm tot}/cH_0$)	$1.37\,(+0.97/-0.54)$

On **every** footing the ratio is consistent with a constant (1.0) and with the DESI decline (0.68), while it **disfavors the rising $cH(z)$ evolution model ($\sim 5\times$) at** $\sim 2\sigma$ (banked posterior probability $\sim$ 1–3%; the exclusion is robust because reaching $5\times$ would require $M_{\rm bar}$ wrong by ~ 0.7 dex, far outside any systematic). This is a real, if narrow, result: one clean object already kills the rising-evolution branch.

What it cannot do is separate the 0.70 decline from flat. The $M_{\rm bar}$ systematic on a single object (≈ 0.3 dex in $M_{\rm bar}$, propagating to ≈ 0.23 dex in the combined a_0 error) is itself larger than the 0.164-dex flat-vs-decline signal at $z = 3.25$. Worse, that 0.226-dex error is a *floor*: it holds the asymmetric-drift coefficient fixed at $\alpha = 3.4$ rather than Monte-Carloing it, whereas varying α moves the ratio from 0.95 ($\alpha = 0$) to 1.27 ($\alpha = 3.4$) — a further ~ 0.1–0.15 dex — so the true error is somewhat larger and the datum even less constraining (the omitted term also pushes a_0 *up*, i.e. anti-framework, so it does not rescue a "win"). We use the anti-framework choices at every fork: the dynamically capped stellar mass, which pushes the ratio *up* (the Monte-Carlo median is 1.31; the capped-mass point value alone is 1.27; the full-SED stellar mass would instead place it at 0.86, right on the DESI decline), and a high asymmetric-drift correction — and the datum still lands consistent with both tracks.

One further systematic works in the framework's favour and is *not* applied to the central value, so the quoted ratio is conservative. The estimator $a_{0,\rm eff} \equiv V^4/(GM_{\rm bar})$ equals a_0 *only* in the deep-MOND limit $g_{\rm bar} \ll a_0$. On the framework's own interpolation $g_{\rm obs}^2 = g_{\rm bar}^2 + g_{\rm bar}a_0$ one has exactly $a_{0,\rm eff} = g_{\rm obs}^2/g_{\rm bar} = g_{\rm bar} + a_0$, so the estimator *overstates* a_0 by $g_{\rm bar}$ whenever the rotation-curve point is not deeply in the MOND regime. For the Big Wheel ($M_{\rm bar} \approx 4.1 \times 10^{11} M_\odot$, outer point at $R \sim 30$–45 kpc) $g_{\rm bar} \approx 0.3$–$0.5\, a_0$, so $a_{0,\rm eff}$ is biased high by ~ 30–50%: subtracting $g_{\rm bar}$ moves the datum from $a_{0,\rm eff} = 1.54 \times 10^{-10}$ (ratio 1.31) toward $a_0 \approx 1.1$–1.2×10^{-10} (ratio ≈ 1.0), i.e. *closer* to flat and to the decline, and *further* from the rise. The "clean deep-MOND" label is therefore optimistic for this single transitional object — the correct treatment uses the full ν-kernel at the measured $g_{\rm bar}/a_0$, not the deep-MOND asymptote — but the direction of the correction only reinforces the "consistent-with-both, rise-excluded" reading, and the actual outer-RC radius (hence $g_{\rm bar}/a_0$) should be pinned before this datum is used quantitatively.

4.3 Why intermediate-z is not a clean a_0 probe

The temptation is to add the many $z \sim 0.5$–2.3 rotating disks (e.g. KMOS$^{\rm 3D}$). They do not help, for a physical reason: these massive rotators sit at $g \gtrsim a_0$, **not** deep-MOND, so their fitted BTFR slopes are 3.0–3.85, not 4, and the zero-point is then degenerate with the slope, the pivot, and baryon/dark-matter-fraction evolution. The inferred "a_0 evolution" *flips sign with the analysis choice*:

- Übler et al. (2017, KMOS$^{\rm 3D}$, fixed local slope 3.75) $\rightarrow$ zero-point ~ 0.3–0.4 dex below local $\rightarrow$ naive map reads a_0 **rising** ~ 2–$3\times$;
- Sharma et al. (2024, free slope 3.21) $\rightarrow$ zero-point above local at their pivot $\rightarrow$ **opposite** sign;
- Di Teodoro et al. (2016) and Tiley et al. (2019, $V/\sigma > 3$) $\rightarrow$ **null**, no significant zero-point evolution.

The scatter across analyses (a factor ~ 2–3, in *either* direction) dwarfs the 0.70-vs-1.0 signal. These points are not valid a_0 readouts; where we include the

$z \approx 2.3$ point in the joint fit we place it at its fixed-slope value 1.86 (rising — the *wrong* sign for the framework decline), so it counts against, not for, the framework.

4.4 The $a_0 = M_{\rm bar}$ degeneracy and the mild distance dependence

Because $\Delta \log a_0 = -\Delta \log M_{\rm bar}$ at fixed V, a mis-estimate of the baryonic mass propagates 1:1 into a_0: the two are algebraically degenerate for this estimator. Rising molecular-gas fractions ($\sim 10\%$ at $z = 0$ to $\sim 50\%$ at $z = 2$), the CO-to-H_2 conversion $\alpha_{\rm CO}$, dust, and IMF evolution all move the BTFR zero-point and are indistinguishable from an a_0 shift with the BTFR / point-mass method. This is the crux limitation of §5. The mild distance–cosmology dependence is separate and quantified: $M_{\rm bar} \propto D^2$, and the ratio $D(\text{DESI})/D(\text{flat})$ shifts $M_{\rm bar}$ by ≤ 0.015 dex at $z = 1$ (smaller at higher z) — roughly $10\times$ below the signal and $15\times$ below the Big-Wheel error. A wrong background moves all points together and cannot manufacture a decline, so the test is **not circular in the fatal sense**: testing whether a_0 tracks $\rho_{\rm DE}$ is not the same as assuming the DE evolution.

4.5 Does the galaxy data show the decline?

No — neither way. Assembling the honest galaxy points (Appendix A, `confront.py`):

set	χ^2(flat)	χ^2(DESI-decl)	$\Delta\chi^2$	lean
all 4 points	1.08	3.05	-1.97	flat, weak
clean deep-MOND only ($z = 0$ + Big Wheel)	0.27	1.55	-1.28	flat, weak

The clean-only row ($z = 0$ + Big Wheel) is the sole formally meaningful statistic; the "all 4 points" row is **illustrative only**, because the intermediate-z points are declared invalid a_0 readouts (§4.3, slope $\neq 4$, sign-contested) and the $z = 1$ entry is itself the median of a hand-picked literature straddle — quantities that cannot legitimately enter a likelihood. Both rows agree in direction: the data mildly prefer flat, $|\Delta\chi^2| \approx 1.3$–$2.0$ — not significant, and what lean exists in the intermediate-z points is toward *rising*, the wrong sign for the framework decline. The distinctive 0.60–0.75 decline is not detected; the data are consistent with both flat and the decline. The one branch the data exclude is the rising $cH(z)$ evolution model ($\sim 2\sigma$, via the Big Wheel).

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5. The key finding: $M_{\rm bar}$ calibration, not rotator count, is decisive

The decisive question for any future test is how the significance scales. The separability statistic — the maximum sigma at which the *current* errors could ever tell flat from the DESI decline — is

$$S = \sqrt{\sum_i \left[\frac{\log a_0^{\rm flat}(z_i) - \log a_0^{\rm DESI}(z_i)}{\sigma_i}\right]^2}.$$

The current data give $S = 0.73\sigma$ (Big Wheel alone), 0.73σ (clean pair $z = 0 + 3.25$), and 0.80σ (all four points): **underpowered**, consistent with both tracks. Every galaxy point sits $\leq 1.2\sigma$ from *both* the flat and the declining track. So more data are needed — but the crucial point is *what kind*.

The naive forecast assumes independent per-object errors that beat down as $1/\sqrt{N}$. The target signal is $-\log_{10}(0.70) = 0.155$ dex, requiring $\sigma_{\rm mean} = 0.052$ dex for 3σ, hence $N \approx 4$ rotators at SPARC-like precision (0.10 dex) or $N \approx 24$ at realistic high-z per-object precision (0.25 dex). **This is wrong**, because the dominant $M_{\rm bar}$ error — $\alpha_{\rm CO}$, IMF, and dust prescriptions applied *identically* to a sample — is largely common-mode, not independent, and a common-mode error does not average down with N. With $\sigma^2_{\rm mean} = \sigma^2_{\rm ind}/N + \sigma^2_{\rm cm}$ and a correlated $M_{\rm bar}$ floor $\sigma_{\rm cm}$:

$M_{\rm bar}$ common-mode floor	N for 3σ
0.00 dex	24
0.05 dex	~ 377
≥ 0.08 dex	**never reaches 3σ at any N**

Because realistic high-z $M_{\rm bar}$ calibration floors are ~ 0.08–0.15 dex ($\alpha_{\rm CO}$ alone is a 0.1–0.2 dex systematic), the test is decisive **only if the common-mode $M_{\rm bar}$ systematic can be driven below ~ 0.05 dex** — a calibration problem, not a counting problem. This is the paper's core methodological contribution: **the decisive observational lever is a uniformly calibrated $M_{\rm bar}$ ladder to $\lesssim 0.05$ dex common-mode, not the number of rotators.** A uniformly calibrated $M_{\rm bar}$ scale is worth more than dozens of noisy objects. (Both footings enter this identically — the forecast is expressed in ratios and dex, so the 20.9% footing choice cancels; it re-enters only the Big-Wheel absolute of §4.2.)

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6. Forecast and the a_0-separable observable

To reach 3σ on the 0.60–0.75 decline versus flat therefore requires (i) a common-mode $M_{\rm bar}$ calibration $\lesssim 0.05$ dex and (ii) a sample of $\sim$ 20–40 clean deep-MOND $z \sim 2$–3 rotators, Big-Wheel-like, with gas-traced outer rotation curves (JWST/ALMA) or individually resolved deep-MOND curves (ELT/HARMONI). For fixed samples the per-object precision required is ≤ 0.16 dex ($N = 10$), ≤ 0.23 dex ($N = 20$), ≤ 0.33 dex ($N = 40$) — all under the assumption that these errors are independent, which returns the argument to the common-mode floor above. Clean deep-MOND disks like the Big Wheel are rare (the intermediate-z KMOS^{3D} rotators are $g \gtrsim a_0$, §4.3), so assembling 20–40 of them is itself an observational stretch.

The deeper methodological escape is to stop using the BTFR zero-point at all. Because a_0 and $M_{\rm bar}$ are 1:1 degenerate in the BTFR, no amount of BTFR data isolates a_0 from baryonic-mass evolution. The **a_0-separable** observable is the *shape* of the rotation curve — the radial-acceleration-relation transition radius / the full RAR — which depends on a_0 at fixed $M_{\rm bar}$ and therefore breaks the degeneracy that the zero-point cannot. A high-z RAR-transition measurement (the acceleration at which the curve departs from the Newtonian-baryonic expectation) reads a_0 directly, without hostage to the gas-mass normalization. This is the observable to target: not more BTFR zero-points, but resolved high-z rotation-curve shapes.

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7. Discussion

The cross-scale link is the substance here. ΛCDM contains no relation forcing a galaxy's internal acceleration scale to equal, or to co-evolve with, the cosmic dark-energy density; the framework forces both. That makes the joint z-tracking of galaxy kinematics and cosmic distances a genuine, falsifiable prediction that is not available to ΛCDM — and it is non-circular precisely because the two datasets never share inputs. The present status is asymmetric between the two scales. On the **cosmology side** the test is live: the DESI DR2 evolving-dark-energy posterior, propagated through the framework relation, produces an $a_0(3)/a_0(0) \simeq 0.74$ decline (at $\sim 2.2\sigma$) that falls inside the framework's 0.60–0.75 band — but this is a consequence of the framework relation plus the DESI $w(z)$, not an independent measurement, and it stands or falls with the DESI signal. On the **galaxy side** the test is a hostage to $M_{\rm bar}$: the distinctive decline is neither detected nor excluded ($S \simeq 0.8\sigma$), the intermediate-z BTFR is systematics-dominated and sign-contested, and the one clean high-z datum can only exclude the rising ALT footing. The honest reading is that the framework's forced bridge is currently underpowered — not passed, not failed — and that the binding constraint is $M_{\rm bar}$ calibration rather than sample size.

Two caveats bound the interpretation. First, the a_0 magnitude inherits the posited Z; only the ratio/tracking is under test, so a detection of the decline would test the *relation*, not the *value* of a_0 or Z. Second, the $z = 0$ tie is definitional and carries no independent weight; the evidential content is entirely in the slope. The framework is honest both ways: at every fork in the Big-Wheel analysis the anti-framework option is taken, and the datum still lands consistent with both tracks; the adverse rising intermediate-z point is retained in the fit; and the forecast is corrected downward by the common-mode floor.

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8. Conclusion

We have set out a non-circular cross-scale test of de Sitter–Unruh modified inertia: the framework replaces ΛCDM's free dark-energy fit with the galaxy acceleration scale, $H^2 = \frac{8\pi G}{3}\rho_m + Z^2 a_0^2(z)/c^2$, forcing $a_0(z)/a_0(0) = \sqrt{\rho_{\rm DE}(z)/\rho_{\rm DE,0}}$ — a bridge between galaxy kinematics and cosmic distances that ΛCDM structurally lacks. The $z = 0$ tie holds at $\sim 1\sigma$ (SPARC $a_0(0) = 1.181 \times 10^{-10}$ vs cosmic $(c/2)\sqrt{G\rho_{\rm DE,0}}$; ratio 1.26 Planck-$H_0$, 1.17 SH0ES-$H_0$), the definitional anchor. The cosmology side is live: the DESI DR2 posterior propagated through the relation yields $a_0(3)/a_0(0) \simeq 0.74$ at $\sim 2.2\sigma$ — a consequence of the framework tie plus the DESI $w(z)$, not an independent detection, and it dissolves if $w \to -1$. The galaxy side is underpowered: the distinctive 0.60–0.75 decline is neither detected nor excluded ($S \simeq 0.8\sigma$); one clean $z = 3.25$ rotator excludes only the ALT-cH_0 rise ($\sim 2\sigma$). The central finding is methodological and redirects the observational strategy: in the BTFR estimator a_0 is 1:1 degenerate with $M_{\rm bar}$, the dominant systematic is common-mode, and with a $\gtrsim 0.08$-dex correlated $M_{\rm bar}$ floor the test never reaches 3σ at any rotator count. The decisive lever is $M_{\rm bar}$ calibration to $\lesssim 0.05$ dex common-mode, and the a_0-separable observable is the full rotation-curve / RAR transition, not the BTFR zero-point. The value of a_0 and the constant Z are posited, not derived; both footings are carried throughout; and the test is reported honestly as a non-detection with a live cosmology-side hint and a strategic redirection of the future program.

Appendix A: The committed, exit-0 verification scripts

All numbers above are reproduced by three committed scripts (numpy/scipy only, exit 0, both footings), re-run 2026-07-18; the frozen public repository is left read-only and all outputs are written to `prep_2026/a0z_crossscale/`.

`galaxy_a0z.py` — the Λ-blind galaxy-side $a_0(z)$ from the high-z BTFR zero-point. Assembles the $z = 0$ SPARC anchor ($1.181 \times 10^{-10} \pm 16\%$), the sign-contested intermediate-z compilation (Übler+2017, Sharma+2024, Di Teodoro+2016, Tiley+2019) with the deep-MOND-validity flag, and the $z = 3.25$ Big-Wheel Monte-Carlo ($a_{0,\rm eff} = 1.54\,(+1.10/-0.61) \times 10^{-10}$). Reports ratios so the footing cancels, and re-enters only the Big-Wheel absolute (SPARC/canonical/ALT = 1.31/1.65/1.37). Prints the forecast $N \approx 4$ (clean) / $N \approx 24$ (realistic) and writes `galaxy_a0z.png`.

`confront.py` — confronts the galaxy $a_0(z)$ against the flat-Λ and DESI-DR2 evolving cosmic tracks. Produces the joint χ^2 (all-4: 1.08 vs 3.05; clean: 0.27 vs 1.55), the separability ($S = 0.73/0.73/0.80\sigma$), the $z = 0$ tie ($1.26/1.44\sigma$ Planck, $1.17/0.95\sigma$ SH0ES), the both-footings Big-Wheel absolute, and the forecast table — the fixed-N precision requirements **and** the common-mode $M_{\rm bar}$ floor of §5 ($\sigma_{\rm cm} = 0 \to N \approx 24$; $0.05 \to N \approx 377$; $\geq 0.08 \to$ never reaches 3σ at any N). Bottom line: $S(\text{all}) = 0.80\sigma \Rightarrow$ underpowered, consistent with both.

`desi_posterior_a0z.py` — propagates the DESI DR2 $w_0 w_a$CDM posterior (three SNe combinations, correlated w_0–w_a via a seeded Cholesky draw) through $a_0(z)/a_0(0) = \sqrt{\rho_{\rm DE}(z)/\rho_{\rm DE,0}}$, giving $a_0(3)/a_0(0) = 0.775$ (Pantheon+, 2.0σ), 0.737 (DESY5, 2.5σ), 0.707 (Union3, 2.1σ); combined median $\simeq 0.74$, decline significance $\sim 2.2\sigma$, all overlapping the framework 0.60–0.75 band. The ratio is footing-, H_0-, and M_B-independent (canonical $\rho_{\rm DE}$ footing by construction; Z cancels).

Supporting reconstructions from an independent lane (`prep_2026/a0z_from_sne/`, `crossscale.py`) confirm that Type Ia supernovae **alone** cannot measure $a_0(z = 3)$: the dark-energy residual $E^2 - \Omega_m(1 + z)^3$ is a difference of large numbers, and the model-independent Gaussian-process reconstruction (Seikel, Clarkson & Smith 2012) is sign-pinned only to $z \approx 0.40$. The robust SNe deliverable is $a_0(0)$ ($\simeq 9.4 \times 10^{-11}$ at $H_0 = 67.4$, $\simeq 1.01 \times 10^{-10}$ at $H_0 = 73.0$), agreeing with SPARC at $\sim 1\sigma$; the $z = 3$ decline numbers come from the DESI/model references, not a SNe reconstruction — which is exactly why the two-dataset (galaxy $\times$ cosmic-distance) construction, and not SNe alone, is the test.

References

Brout, D., et al. 2022, *ApJ* **938**, 110 (Pantheon+).

DESI Collaboration 2025, arXiv:2503.14738 ($w_0 w_a$CDM; ΛCDM excluded at $2.8/3.8/4.2\sigma$).

Di Teodoro, E. M., Fraternali, F., & Miller, S. H. 2016, *A&A* **594**, A77.

Lelli, F., McGaugh, S. S., & Schombert, J. M. 2016, *AJ* **152**, 157 (SPARC).

McGaugh, S. S. 2005, *ApJ* **632**, 859; McGaugh, S. S. 2012, *AJ* **143**, 40 (BTFR).

Milgrom, M. 1983, *ApJ* **270**, 365; Milgrom, M. 1999, *Phys. Lett. A* **253**, 273 (the ν-kernel, Eq. 9; astro-ph/9805346).

Seikel, M., Clarkson, C., & Smith, M. 2012, *JCAP* **06**, 036 (Gaussian-process $H(z)$).

Sharma, G., Upadhyaya, V., Salucci, P., & Desai, S. 2024, *A&A* **690** (Tully–Fisher at $0.6 \leq z \leq 2.5$; free bTFR slope 3.21 ± 0.28; arXiv:2406.08934).

Tiley, A. L., et al. 2019, *MNRAS* **482**, 2166 (KROSS–SAMI Tully–Fisher across 8 Gyr since $z \approx 1$).

Übler, H., et al. 2017, *ApJ* **842**, 121 (KMOS^{3D} bTFR).

Wang, W., et al. 2025, *Nature Astronomy* **9**, 710 (the "Big Wheel" giant disk at $z = 3.2514$; arXiv:2409.17956).

Zimmerman, C. 2026, *Reading the Cosmological Constant from Dwarf-Galaxy Rotation Curves: The a_0-Line* (self-cite), DOI 10.5281/zenodo.21419735.

Zimmerman, C. 2026, *MI Field Theory Results 2026* (self-cite), DOI 10.5281/zenodo.21403470.